

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

WE DO examples

- Ex 1: Gold Coast
- Ex 2: Uluru
- Ex 3: 15–24 years
- Ex 4: 12%
- Ex 5: Jan and Feb, 18°C
- Ex 6: e.g. fewer people chose “no religion”

Set A

- Gold Coast
- Uluru
- 30
- true (8)

Set B

- 15–24 years
- 12%
- 16.5%
- e.g. working-age adults and migrants

Set C

- Jan and Feb, 18°C
- e.g. Jan and Feb (or May and Sep)
- October
- July (3°C)
- May–June or June–July (2°C)
- it falls steadily to a low in July

Set D

- 22%
- Anglican
- No religion
- e.g. more Christians and fewer “no religion”

Extension

- 76.5°
- Gold Coast 6 cm, Uluru 3.4 cm, Snowy 3 cm, Kangaroo Is. 2 cm, Phillip Is. 1.6 cm

Exit ticket

- Level 1: Phillip Island
- Level 2: 11°C
- Level 3: 8%

10.02 Misleading graphs (booklet p. 7)

WE DO examples

- Ex 1: the vertical axis starts at \$2 million, not 0
- Ex 2: profits grew much faster than they did
- Ex 3: no
- Ex 4: 51 and 56
- Ex 5: uneven scale, pictures, no labels, missing data
- Ex 6: to impress investors

Set A

- axis does not start at 0
- huge growth
- \$2.2 million
- \$1.8 million

Set B

- 50
- 51
- 58
- no

Set C

- line graph from 0 to 4 with even scale

Set D

- the area is 4 times bigger, not 2
- you do not know what is shown
- the steps are not equal
- own answer

Extension

- the axis starts at 2, so heights above 2 are compared (0.2 vs 2.0)
- e.g. axis from 0 to 20

Exit ticket

- Level 1: e.g. the axis does not start at 0
- Level 2: small differences look huge
- Level 3: start the vertical axis at 0 with an even scale

10.03 Dot plots (booklet p. 11)

WE DO examples

- Ex 1: 23°C
- Ex 2: 15°C
- Ex 3: 4
- Ex 4: $\frac{1}{15}$
- Ex 5: dot plot
- Ex 6: most days between 20°C and 26°C, peak at 23°C

Set A

- 23°C
- 15°C
- 4
- $\frac{1}{15}$

Set B

- 16
- 1
- 7
- 8

Set C

- dot plot
- dot plot

Set D

- dot plot
- mode 0, outlier 15; it makes the mean bigger

Extension

- 18.8%
- still 1

Exit ticket

- Level 1: a graph with one dot for each value on a number line
- Level 2: 1
- Level 3: 8

10.04 Stem-and-leaf plots (booklet p. 15)

WE DO examples

- Ex 1: 25
- Ex 2: 71 kg
- Ex 3: 80 kg
- Ex 4: 106 kg
- Ex 5: 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
- Ex 6: to see the shape and find the median and mode easily

Set A

- 25
- 71 kg
- 80 kg
- 106 kg

Set B

- 15
- 45
- 75
- 66

Set C

- 2 | 1 3 7 8; 3 | 0 2 5 6 9; 4 | 1 4 5
- 6 | 4 7 8; 7 | 0 3 5 8 9; 8 | 2 5 8; 9 | 1

Set D

- 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
- 9 players, 9/25

Extension

- 10 | 5; 11 | 2 8; 12 | 1 5; 13 | 0
- 25 hundredths (0.25)

Exit ticket

- Level 1: 6
- Level 2: 40, 42, 47
- Level 3: 1 | 2 4 7; 2 | 1 5 9

10.05 Mean and mode (booklet p. 19)

WE DO examples

- Ex 1: 4
- Ex 2: 4
- Ex 3: 8.78
- Ex 4: 6
- Ex 5: 3.5; 1
- Ex 6: 6

Set A

- 4
- 6
- 1
- 7
- 12
- red

Set B

- 5
- 4
- 30
- 5
- 3.5
- 9

Set C

- 8.78; 6
- 3.5; 1
- 14; 15
- 4.07; 3.5
- 49.86; 52
- 1.88; 2

Set D

- 6
- 84
- e.g. 8, 8, 10, 11, 13
- yes, e.g. 1, 1, 2, 2 (modes 1 and 2)

Extension

- 10.4
- it is a balance point, e.g. mean of 1 and 2 is 1.5

Exit ticket

- Level 1: 5
- Level 2: 8
- Level 3: 6.5

10.06 Median and range (booklet p. 23)

WE DO examples

- Ex 1: 4
- Ex 2: 4
- Ex 3: 10
- Ex 4: 6
- Ex 5: 3.5; 7
- Ex 6: the mean

Set A

- 4
- 6
- 7
- 19
- 13
- 8

Set B

- 8
- 4
- 5
- 7
- 6
- 22.5

Set C

- 10; 6
- 3.5; 7
- 14.5; 6
- 3.85; 1.6
- 52; 23
- 4; 98

Set D

- mean 7, median 3.5, mode 3, range 23
- the median: the outlier 25 pulls the mean up
- e.g. 1, 4, 6, 8, 11
- 57

Extension

- 3, 5, 9, 9
- a few very high salaries make the mean misleading

Exit ticket

1. Level 1: 6
2. Level 2: 25
3. Level 3: 9

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 27)

WE DO examples

1. Ex 1: 7
2. Ex 2: 1.5
3. Ex 3: 66
4. Ex 4: 64
5. Ex 5: 61.2
6. Ex 6: mostly in the 60s

Set A

1. 7
2. 1.5
3. 1
4. 1.88

Set B

1. 30
2. 64
3. 66
4. 61.2

Set C

1. 13
2. 23
3. 23
4. 23.2
5. 23.48
6. the range

Set D

1. 35
2. 83
3. 80
4. 85.88

Extension

1. the median: it is not affected by extreme values
2. 1.53

Exit ticket

1. Level 1: 3
2. Level 2: 53.5
3. Level 3: 18

10.08 Comparing data (booklet p. 31)

WE DO examples

1. Ex 1: the marks of the girls (left) and boys (right)
2. Ex 2: 94, a girl
3. Ex 3: 54
4. Ex 4: 83
5. Ex 5: girls: mean 55, median 54; boys: mean 55.8, median 60
6. Ex 6: girls: range 76 vs boys 83

Set A

1. 94, girl
2. 10, boy
3. 8
4. 13

Set B

1. 55
2. 54
3. 76
4. 55.8
5. 60
6. 83

Set C

1. Class A: 5 | 8 2, 4 | 5 ...; Class B on the right
2. Class A: higher median, smaller range

Set D

1. boys have a slightly higher median (60 vs 54) but a larger range
2. girls: smaller range (76 vs 83)
3. median, as there are some very low marks
4. own questions

Extension

1. Ana: she scores 12 or more every game
2. e.g. 5, 5, 5 and 0, 5, 10

Exit ticket

1. Level 1: comparing two groups
2. Level 2: A
3. Level 3: A